Rachel Youm

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Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL, HTML/CSS
Frameworks: FastAPI, Node.js, Express.js, Three.js, React Three Fiber, React, SQLAlchemy, Alembic
Tools: PostgreSQL, MongoDB, Redis, Docker, Git, Linux, Bash, GCP, Vite, Vercel, Railway

Education

Simon Fraser University — Burnaby, BC
Bachelor of Science (BSc), Computer Science — *Aug. 2022 – Expected Apr. 2028*

Projects

Mapsi — AI Travel Planner (team of 4) — StormForge 2026
- Implemented **40+ FastAPI REST endpoints** backed by PostgreSQL for an AI-powered travel planning platform
- Designed and built an AI itinerary pipeline using OpenAI and Google Places to generate personalized travel plans
- Modeled PostgreSQL schemas with **SQLAlchemy/Alembic** and enforced owner, editor, and viewer permissions
- Cut response times from **239 ms to 90 ms** with **Redis caching** using 30-minute to 24-hour TTLs
- Containerized backend with Docker and deployed to Google Cloud Run for scalable development

Prioritize (team of 5) — Sep 2025 – Dec 2025
- Built a weighted scheduling engine that automatically prioritized tasks using deadlines, difficulty, and grade weight
- Implemented **15+ FastAPI REST endpoints** backed by PostgreSQL for task management and smart scheduling
- Implemented JWT authentication with bcrypt to support secure multi-user access and protected API endpoints
- Integrated Google Calendar API and ICS export for seamless calendar synchronization
- Developed Redis-backed event logging APIs to support real-time user activity analytics

SweatMarket — Hackathon – Sep 2025
- Built a full-stack fitness platform with **FastAPI, SQLModel, and PostgreSQL** for creating and joining workouts
- Engineered a **QR + geolocation check-in system** with signed tokens, proximity checks, and time validation
- Developed real-time group chat with **WebSockets**, participant-only access, image uploads, and live updates
- Built **36** Pytest/TestClient tests with **79%** backend statement coverage; automated Docker/GitHub Actions CI
- Integrated **Google Places and browser geolocation APIs** for workout search and location verification

Experience

Barista — Dec 2024 – Oct 2026
Paris Baguette — *Burnaby, BC*
- Coordinated with teammates during peak hours to manage concurrent orders and maintain accurate, timely service

Math Tutor — Sept 2024 – Dec 2025
Private Tutor — *Coquitlam, BC*
- Tutored 4 students in calculus, helping 2 improve their grades by at least one letter grade

Barista & Server — Sept 2023 – Sept 2024
Ibex Cafe + Kitchen — *Coquitlam, BC*
- Managed concurrent customer requests and adapted to changing priorities while coordinating service with teammates